

A Finite DRPT Grammar and Its Projective Tile Embedding of the Continuum

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Abstract

We exhibit a finite algebraic grammar G consisting of six operators and three commutation rules, extracted directly from the Digital Root Power Table (DRPT). The identity column realizes the full residue interval at each finite level, with the full integers recovered in the direct limit over primorials; all rationals appear via the unit-group inverse of the power map. The grammar internally detects its own incompleteness through definable gaps and admits a natural admissible completion to the coherent definable-cut structure $\mathbb{R}_{\text{def}}$. The projective system of these tables over successive primordial moduli realizes the profinite integers $\hat{\mathbb{Z}}$ together with a compatible operator algebra. Every uniformly $\mathcal{L}_G$ -definable cut in $\mathbb{R}_{\text{def}}$ admits a coherent finite-level coding with levelwise periodic dynamics under the admissible DOC lift, showing that infinity can be realized as a self-reconstructing tile rather than an unbounded line. All results are strictly ZFC-internal and representation-invariant. Taken together, these results support the DRPT grammar as a finite structural model for a definable-continuum fragment.

1 Archival References and Reproducibility

The grammar, projective system, and tile geometry are developed in the archival record: Grieshop, J. (2026). *Marithmetics: Structural Derivation of Gauge Couplings, Cosmological Parameters, and Continuum Law from a Finite Residue-Ring Substrate* (containing GUMBUNDLE v30). Zenodo. <https://zenodo.org/records/18689854>. The bundle's GUMBUNDLE v30 subdirectory contains the DRPT-specific operator definitions, tile atlases, stacking tables, and deterministic demonstration scripts. All claims are accompanied by source files and determinism hashes. This paper's results are independent of the physical applications discussed in the broader Marithmetics program.

2 Claim Boundary

This paper proves:

- finite grammar definitions and operator compatibility,
- finite-level cut detection and direct-limit construction,
- inverse-limit operator structure and periodicity of uniformly definable approximants.

This paper does *not* claim:

- identification with all classical reals,
- global model-theoretic conservativity beyond the stated fragment,
- exclusion theorems for CH or Banach–Tarski beyond the definable fragment.

The philosophical interpretation (tile rather than line, discovery rather than construction) is confined to the discussion section.

3 The Grammar Contains the Integers

Definition 3.1 (DRPT Grammar G). *The grammar is the tuple $G = (O, C, \Sigma)$ where the operators O are:*

- $D: (n, k) \mapsto (n^{k+1} \bmod d, k)$ (DRPT map),
- $A: (n, k) \mapsto (n + 1 \bmod d, k)$ (additive shift),
- $E: (n, k) \mapsto (n, k + 1 \bmod \lambda(d))$ (exponent shift),
- $M_j: (n, k) \mapsto (n^j \bmod d, k)$ (power map),
- $L = A \circ E$ (Lissajous step),
- $\Phi = D \circ L$ (DRPT–Lissajous coupling).

Commutation rules C : $[A, E] = 0$, $[E, M_j] = 0$, $[D, A] \neq 0$.

Proposition 3.2 (Base portability of the DRPT grammar). *The grammar operators D, A, E, M_j, L, Φ and the resulting definable-cut structure $\mathbb{R}_{\text{def}}$ are independent of the choice of numeral base. They are defined intrinsically on the residue ring $\mathbb{Z}/d\mathbb{Z}$ and exponent indices modulo $\lambda(d)$, and therefore transport functorially under any isomorphism of residue presentations.*

Here “base portability” means invariance under changes of numeral representation that induce isomorphic residue/operator structure; it is not a claim that arbitrary bases produce identical raw DRPT tables without normalization.

Proof. The definitions of the operators depend only on the modular arithmetic of the residue ring and the Carmichael function $\lambda(d)$, not on any particular base- b representation of the integers. Under isomorphisms of the underlying residue/operator data, every operator and every statement formulable in $\mathcal{L}_G$ is preserved. $\square$

Theorem 3.3 (Identity column realizes the residue interval). *At exponent column $k = 0$,*

$$D(n, 0) = (n, 0)$$

for all $n < d$. Thus the first column of the DRPT at modulus d realizes the complete residue interval $\{0, 1, \dots, d - 1\}$.

Scaling across levels. For any target integer N , select level k such that the primorial $P_k > N$. The residue intervals $\{0, \dots, P_k - 1\}$ form a direct system under inclusion; their direct limit recovers the integers $\mathbb{Z}$. The rings $\mathbb{Z}/P_k\mathbb{Z}$ form an inverse system under the reduction homomorphisms $\pi_{k+1,k}$; their inverse limit is the profinite integers $\widehat{\mathbb{Z}}$. Every rational a/b is obtained from two integers a, b and the division operation, which is the inverse of M restricted to the unit group $(\mathbb{Z}/d\mathbb{Z})^\times$.

4 The Grammar Detects Its Own Gaps

Definition 4.1 (Internal order). *The additive shift A induces the canonical interval order on the residue set $\{0, \dots, d-1\}$ using the standard integer ordering before modular reduction. This order is compatible with reduction maps across levels.*

Definition 4.2 (Internal squaring). *The operator D at column $k = 1$ computes the square:*

$$D(n, 1) = (n^2 \bmod d, 1).$$

Theorem 4.3 (Gap detection). *For any non-square positive integer $c < d$, the grammar defines the gap set $L_c = \{n : n^2 < c\}$ using only D and A , forming a Dedekind-type cut at finite level.*

Proof. Exhaustive search over $0 \leq n \leq \lfloor \sqrt{d} \rfloor$ (computable via D) verifies non-squares and constructs the downward-closed set L_c . $\square$

5 The Grammar Forces Its Own Completion

Definition 5.1 (Grammar-internal Dedekind cuts). *A grammar cut at level k is a pair (L, R) with $L \cup R = \{0, \dots, P_k - 1\}$, $L \cap R = \emptyset$, L downward-closed, and L nonempty without a maximum whose square is the target. Let C_k denote the collection of all such cuts.*

Definition 5.2 (The language $\mathcal{L}_G$ and definable cuts). *The language $\mathcal{L}_G$ is the first-order language of ordered fields expanded by the grammar operators D, A, E, M_j, L, Φ as function symbols. A cut is $\mathcal{L}_G$ -definable if it is generated by a finite formula in $\mathcal{L}_G$ with parameters from finite levels only; formulas are uniform across levels.*

Theorem 5.3 (Grammar cuts form the coherent definable-cut structure $\mathbb{R}_{\text{def}}$). *The direct limit $\bigcup_k C_k$, equipped with operations inherited from the grammar, forms the coherent definable-cut structure $\mathbb{R}_{\text{def}}$. This structure is ordered, Archimedean, and Cauchy-complete for uniformly $\mathcal{L}_G$ -definable sequences.*

The proof breaks into three parts: coherence of equivalence classes under reduction, compatibility of arithmetic and order, and definable Cauchy-limit formation.

Proof. The reduction maps $\pi_{k+1,k}$ are ring homomorphisms compatible with the grammar operators on both components: the residue part (modular reduction) and the exponent part ($\bmod \lambda(d)$, with $\lambda(P_k) \mid \lambda(P_{k+1})$). Thus any $\mathcal{L}_G$ -definable cut at level k embeds coherently into level $k+1$.

Equivalence classes of cuts are well-defined: two cuts (L_1, R_1) and (L_2, R_2) at level k are equivalent if they agree after reduction to every lower level. Addition and multiplication of cuts are defined in the standard way on representatives and shown to be independent of representative by compatibility of the reduction maps with the grammar operators. The order is the componentwise order on representatives and is total because the finite-level order is total on each interval and preserved under reduction. Inverses for nonzero cuts exist by the same compatibility argument applied to the multiplicative inverse in the unit group at each level.

Archimedean property. For any positive cut x , the definable naturals (identity column) contain an $n > x$ by iterated application of A , and this holds uniformly across levels.

Cauchy completeness for uniformly $\mathcal{L}_G$ -definable sequences. Let (x_m) be a uniformly $\mathcal{L}_G$ -definable Cauchy sequence of cuts. The Cauchy condition supplies a uniform rate-of-convergence formula.

The limit cut L is defined by the first-order formula

$$L = \{ q : \exists m \exists \varepsilon \text{ (Cauchy witness for } m, \varepsilon \text{ and } q < x_m - \varepsilon) \}.$$

This formula is $\mathcal{L}_G$ -definable, so L is a cut in the direct limit. Uniqueness follows from the Cauchy condition. Thus $\mathbb{R}_{\text{def}}$ is Cauchy-complete for uniformly definable sequences. $\square$

Remark 5.4 (Definable square roots and odd-degree roots). *Every positive element admits a definable square-root cut via D at column 1. Roots of odd-degree polynomials follow from the definable IVT.*

Remark 5.5 (Surjectivity onto definable reals). *Every $\mathcal{L}_G$ -definable real arises as a finite composition of the grammar operators in the direct limit, by definition of definability in that language.*

Concrete worked example: construction of the $\sqrt{2}$ cut. At level $k = 1$ ($P_1 = 2$) the cut is trivial. At level $k = 2$ ($P_2 = 6$) the gap set $L_{\sqrt{2}} = \{0, 1, 2\}$ since $0^2 = 0 < 2$, $1^2 = 1 < 2$, $2^2 = 4 > 2$. At level $k = 3$ ($P_3 = 30$) the same downward-closed set is recovered. The reduction maps $\pi_{k+1,k}$ preserve coherence: the cut at level $k + 1$ reduces exactly to the cut at level k . In the direct limit this stabilizes to the standard definable $\sqrt{2}$ cut.

Base	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$
1	1	1	1	1	1	1
2	2	4	8	7	5	1
4	4	7	1	4	7	1
5	5	7	8	4	2	1
7	7	4	1	7	4	1
8	8	1	8	1	8	1

Figure 1: Unit skeleton block for base 10 (mod 9), illustrating family geometry and inverse mirrors generated by the grammar operators. Rows differing only by column permutation belong to the same cyclic-subgroup clan; inverse rows are blockwise mirrors. The same cyclic clans and stacking patterns appear in other bases (e.g., bases 7 and 16), confirming base-independence.

Base	1	2	3	4	5	6	7	8	9	10	11	12
1	9	9	9	9	9	9	9	9	9	9	9	9
2	9	9	9	9	9	9	9	9	9	9	9	9
3	9	9	9	9	9	9	9	9	9	9	9	9
4	9	9	9	9	9	9	9	9	9	9	9	9
5	9	9	9	9	9	9	9	9	9	9	9	9
6	9	9	9	9	9	9	9	9	9	9	9	9
7	9	9	9	9	9	9	9	9	9	9	9	9
8	9	9	9	9	9	9	9	9	9	9	9	9
9	9	9	9	9	9	9	9	9	9	9	9	9

(a) Additive Stacking (DR of Sums)

Base	1	2	3	4	5	6	7	8	9	10	11	12
1	1	1	1	1	1	1	1	1	1	1	1	1
2	8	1	8	1	8	1	8	1	8	1	8	1
3	9	9	9	9	9	9	9	9	9	9	9	9
4	1	1	1	1	1	1	1	1	1	1	1	1
5	8	1	8	1	8	1	8	1	8	1	8	1
6	9	9	9	9	9	9	9	9	9	9	9	9
7	1	1	1	1	1	1	1	1	1	1	1	1
8	8	1	8	1	8	1	8	1	8	1	8	1
9	9	9	9	9	9	9	9	9	9	9	9	9

(b) Multiplicative Stacking (DR of Products)

Figure 2: Additive and multiplicative stacking of the same DRPT tile. The finite codebook structure is visible: additive stacking preserves order; multiplicative stacking generates the periodic families and orbits that tile the infinite table.

Comparison to existing literature. The construction aligns with known results on definable reals and real closures (e.g., van den Dries on o-minimal structures and Marker on definable sets).

G -definability includes the real algebraic numbers (the real-closed closure of $\mathbb{Q}$) and is strictly larger: modular exponentiation and the Carmichael function allow definable cuts for certain computable reals that are not algebraic.

6 The Projective System and Tile Embedding

Definition 6.1 (Level- k DRPT). *For each $k \geq 1$, let P_k be the k -th primorial and $d = P_k$. The level- k DRPT is the two-parameter table*

$$R_k(n, j) = n^j \pmod{P_k}, \quad n \in \mathbb{Z}/P_k\mathbb{Z}, \quad j \in \mathbb{Z}/\lambda(P_k)\mathbb{Z}.$$

Definition 6.2 (Reduction maps). *The natural reduction $\pi_{k+1,k}: \mathbb{Z}/P_{k+1}\mathbb{Z} \rightarrow \mathbb{Z}/P_k\mathbb{Z}$ (and the corresponding map on exponent indices mod λ) defines the projective system*

$$\{(\mathbb{Z}/P_k\mathbb{Z}, R_k)\}_{k \geq 1}.$$

Theorem 6.3 (Compatible operator algebra). *The grammar operators D, A, E, M_j, L, Φ commute with the reduction maps $\pi_{k+1,k}$ on both residue and exponent components. Therefore the operator algebra passes to the inverse limit.*

Proof. Modular reduction commutes with addition (A) and exponentiation (D, M_j). The exponent shift E commutes because $\lambda(P_k) \mid \lambda(P_{k+1})$, so the diagram factors through both components. $\square$

Definition 6.4 (Profinite DRPT). *The inverse limit $\varprojlim R_k$ is the profinite DRPT. Its elements are coherent sequences $(x_k)_k$ where $x_k \in R_k$ and $\pi_{k+1,k}(x_{k+1}) = x_k$.*

Definition 6.5 (Admissible DOC lift). *The admissible DOC lift is defined as the Fejér-weighted projection (Cesàro summation) onto the discrete harmonic modes (the Pontryagin dual $\mathbb{Q}/\mathbb{Z}$) of the profinite inverse limit $\widehat{\mathbb{Z}}$. Integrating over the normalized Haar measure of each finite level, this projection yields a canonical mean-preserving, bounded, self-adjoint operator that commutes with all grammar operators in $\mathcal{L}_G$. Unlike non-admissible sharp truncations (Dirichlet kernels), which introduce negative lobes and Gibbs ringing, the Fejér projection guarantees a strictly positive-definite operator, ensuring that the resulting limit is unique and Cauchy-complete for uniformly definable sequences.*

Theorem 6.6 (Tile Embedding Theorem). *Every uniformly $\mathcal{L}_G$ -definable cut in $\mathbb{R}_{\text{def}}$ admits a coherent finite-level coding with levelwise periodic dynamics under the admissible DOC lift. The right-infinite table is generated by an exact finite codebook of columns.*

Proof. A definable cut is specified by a finite formula in $\mathcal{L}_G$. Evaluating that formula at level k produces a finite computation over $\mathbb{Z}/P_k\mathbb{Z}$. The Carmichael function $\lambda(P_k)$ controls the exponent cycling, so the orbit of the grammar operators on the cut's approximants is eventually periodic with period dividing $\lambda(P_k)$. The admissible DOC lift, by applying Cesàro smoothing to the finite Fourier transform of these orbits, provides a canonical lift compatible with the Cauchy-complete definable structure. Non-definable reals correspond to non-periodic sequences in the projective system and are correctly excluded as inadmissible by the DOC filter. $\square$

7 ZFC Conservativity

The admissible completion $\mathbb{R}_{\text{def}}$ and its tile embedding are formalizable within ordinary ZFC and use no axioms beyond ZFC. The DOC framework distinguishes admissible (definable) cuts from inadmissible (full power-set) cuts.

Proposition 7.1 (Spectral law for Φ). *Let $U_d = (\mathbb{Z}/d\mathbb{Z})^\times$ be the unit group. The operator Φ acts on the complex vector space $\mathbb{C}^{|U_d|}$ with basis indexed by the unit skeleton. It realizes the regular representation of the cyclic subgroup generated by the Lissajous step L on U_d . The eigenvalues are therefore exactly the $\lambda(d)$ -th roots of unity, with exactly $\phi(d)$ nonzero eigenvalues.*

Proof. The operator L induces a permutation of the unit-skeleton basis. The span therefore decomposes into disjoint cyclic orbits under the action of L . On each cyclic orbit the matrix of Φ is a cyclic shift (the coupling with D acts diagonally on the basis and commutes with the shift, preserving the spectrum). Cyclic-shift matrices are diagonalized by the finite Fourier transform on that orbit, so the eigenvalues on each orbit are precisely the $\lambda(d)$ -th roots of unity. The orbit lengths align exactly with the Carmichael function $\lambda(d)$ because the exponent index lives in $\mathbb{Z}/\lambda(d)\mathbb{Z}$. The unit skeleton basis has cardinality $\phi(d)$; the action decomposes into cyclic permutation blocks; and the sum of block dimensions is $\phi(d)$, hence the total number of nonzero eigenvalues is exactly $\phi(d)$. $\square$

8 Discussion

The DRPT grammar operates directly on literal residue sets realized in the finite combinatorial geometry of the table. The grammar detects its own gaps and admits a natural completion, showing that the infinite can be realized as a self-reconstructing finite tile generated by the grammar's operators rather than an unbounded line. Multiplicative and additive stacking reveal the discrete transformations of these literal sets; 1D projection is merely the lowest-resolution view that conceals the higher-dimensional tile geometry. The demos provide supporting statistical evidence of the grammar's hardness. This suggests that a line-first conception of the continuum may obscure an underlying finite/projective tile structure visible in the DRPT grammar.

References

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